Transformer-Based Band Regrouping With Feature Refinement for Hyperspectral Object Tracking

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***Abstract*— Hyperspectral videos (HSVs) offer not only spatial information but also diagnostic spectral features. Due to the fact that spectral features are only related to the material of the object, this advantage can address the issue of RGB video tracking failure when the object and background are visually similar. However, the effectiveness of deep learning models is limited due to insufficient HSV training data. Existing methods tend to divide a hyperspectral image (HSI) into several three-channel false-color images to leverage the existing RGB trackers for transfer learning. Nonetheless, these methods lack adequate exploration of band interrelations and overlook correlation among objects prior to similarity calculation. In this article, a transformer-based band regrouping and feature refinement network (TBR-Net) is introduced, which is specifically tailored for hyperspectral object tracking. To maximize the potential of the RGB tracker and enhance the use of available training data, we propose a transformer-based band regrouping (TBR) method. By modeling long-range spectral dependencies, the inherent context information among bands is captured, which is subsequently utilized to reorganize bands into several false-color images. Furthermore, to combine the relationship of the template and the search (T & S) frames into a correlation calculation, a feature refinement module (FRM) is designed. The cross-attention mechanism enables mutual relation modeling, allowing similar regions to be perceived and form discriminative feature representation. As a result, a hyperspectral tracker can be efficiently trained via transfer learning to address the data insufficiency challenge, while the mutual perception between objects further enhances the tracking performance. Its effectiveness is validated by extensive benchmark experiments, which demonstrate that the TBR-Net surpasses state-of-the-art methods.**

***Index Terms*— Band selection, deep learning, hyperspectral object tracking, visual transformer.**

I. Introduction

VISUAL object tracking, particularly single-object tracking, constitutes a vital subdiscipline within computer vision fields. It employs a tracker to anticipate the movement of an object across successive frames using the object's initial position from a given image. Object tracking is widely used in security surveillance [2], automatic driving [3], and other fields [4], [5], [6], [7]. Due to the convenience of data acquisition and the wide range of application scenarios, most of the existing research focuses on processing RGB three-channel images. Benefiting from the rich color information, RGB-based methods can extract visual features, compute the similarities between objects, and subsequently track identical objects. However, when an object and its background share visual similarities, it can confound the tracker and impede the tracking performance [8], [9]. As shown in Fig. 1, objects in the same color as the background can lead to significant visual interference.

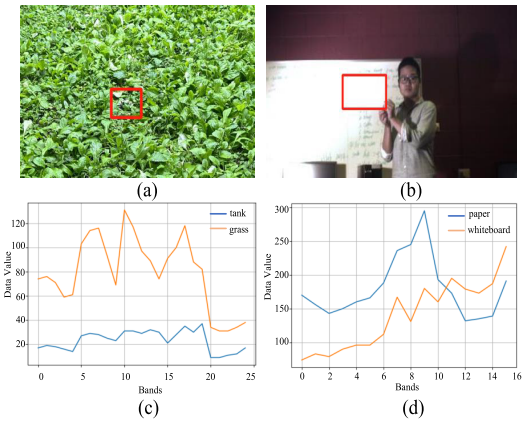

Fig. 1. Illustration of different objects in the same color but different spectral curves. (a) Shows a plastic model (red-framed) of a tank placed on the grass (captured by the 25-band snapshot mosaic HS camera). (b) Shows a white paper (red-framed) placed in front of a whiteboard (HOTC dataset [1]). (c) and (d) Demonstrate the corresponding spectral image pixel values for each band, representing the reflectance to the wavelength. It can be seen that objects in the same color contain different spectral information.

Hyperspectral image (HSI) acquisition has become easier with the development of hyperspectral (HS) snapshot cameras [10], [11], [12]. The HS technology can simultaneously obtain the spatial and spectral information of objects [1]. In this way, the HS snapshot cameras can accurately distinguish objects with the same visible but overall different spectral features, making super-resolution [12], classification [13], and tracking [1] of objects possible.

In the earlier studies, the majority of research primarily relied on handcrafted features to exploit the extensive spectral information of HSIs. Xiong et al. [1] constructed an HS video dataset for object tracking and proposed a new method to capture the 3-D spectral with spatial structures of an HSI by

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the spectral–spatial histogram of multidimensional gradients. Zhang et al. [14] proposed a channel selection strategy to select three bands, which are fed to the network to predict the location of objects. Hou et al. [15] designed a regularizer to refine the spectral features corresponding to the center pixel of the correlation filters. With the development of deep learning, convolutional features are more discriminative compared with handcrafted features. Li et al. [16] designed a band attention block to generate band weights and used the band weights to generate three-channel images. In this way, the ensembled color tracker can be used to tackle the challenge of HSI tracking. Liu et al. [17] proposed an HSI tracker based on the RGB tracker, with the information fused from spatial and spectral features to tackle the "data insufficiency" problem. Ouyang et al. [18] proposed a transfer learning framework to group the bands into several groups to leverage the ability of RGB trackers. Li et al. [19] proposed an autoencoder-like self-expressive (SE) model to reconstruct the spectral bands and to capture their intrinsic relationship. In this way, the HSI cubes can be regrouped into several three-channel false-color images.

Based on this, the pretrained RGB trackers can be used to fill the gap of HSI data scarcity and fine-tune the HS trackers. Most existing deep learning-based methods are typically split into two steps: band grouping and transfer learning using the existing RGB trackers on HSIs. However, the current band grouping methods, particularly those based on the channel attention mechanism—which generate spectral weights in the form of fully connected layers—tend to overlook the modeling of long-range spectral dependencies between bands, thereby limiting the discrimination of the three-channel false-color images. Furthermore, the fine-tuning process from the existing RGB tracker often neglects the spatial relationship between the template and the search (T & S) frames.

Recently, the Transformer has been widely used for object tracking due to its potential relational modeling capabilities. For the visual objects tracking, Chen et al. [20] proposed a new Transformer-based feature fusion network that alters the traditional method of computing the relationship between template frames and search frames by utilizing self-attentive and transattentive modules to extract global contexts. Wang et al. [21] bridged the template frames through the encoder and search frames through the decoder of the Transformer. For the HSI tracking, Zhao et al. [22] utilized the discriminative information from RGB and HS modalities jointly, mining spatial and spectral features of data to improve tracking performance. Despite the potential advantages, the utilization of the Transformer in the field of HSI tracking is less common, and the existing networks often result in inadequate exploration of relationships between spectral bands and between templates and search frames.

To address these limitations and leverage the inherent relationship modeling capabilities of the Transformer, we design a novel HS tracking framework to maximize the extraction of relevant information from HS data for improved object tracking. Specifically, a Transformer-based band regrouping and feature refinement network (TBR-Net) is proposed. TBR-Net includes three components: a Transformer-based band regrouping (TBR) method, a backbone network, and a tracking head with a feature refinement network. First, self-representation-based subspace clustering (SSC) models play an important role in band selection, assuming that each HS band can be inter-represented by other bands through a linear transformation [23]. The core of the SSC is to learn the coefficient matrix through an SE model, which reveals the intrinsic relationship between spectral bands [24]. However, the assumption of linear relationship brings the following shortcomings.

1) It is weak for the linear assumption to capture the inherent global information of the dictionary (band sequences) [25].
2) The self-representation adopts the original spectral cube to generate the importance coefficient, neglecting the correlation of high-level semantic information [26]. Thus, TBR is designed to learn the coefficient matrix, which represents the importance of each band in contributing to representation quality through contextual relation modeling. With the mining ability of long-range latent relation, the Transformer encoder can capture the multiscale local and global information, which suppresses the ineffective spectral band and focuses on the more robust pattern that contributes to tracking performance in end-to-end training. Furthermore, the existing trackers treat T & S frames independently in the correlation modeling, limiting the tracking performance. Considering the cross-attention mechanism in terms of region of interest (ROI) focus property, a feature refinement module (FRM) is presented, where the pixel-level similarity between T & S frames is formulated with respect to different linear transformations. Based on the cross-attention weights, the TBR-Net can focus on patches in the search image that are similar to the template image.

The main contributions of this article are as follows.

1) A TBR method is proposed to capture the spectral context information and regroup the bands to several three-channel images with informative combination. This helps to identify and prioritize the most informative bands, thereby improving the tracking performance.
2) An FRM is proposed to model the similarity between pixels in terms of ROI focus property. The relevance of T & S frames can be brought to the existing cross-correlation calculation to generate a more robust feature representation.
3) Extensive experiments with many state-of-the-art trackers are conducted to demonstrate the effectiveness of the proposed TBR-Net.

The remaining structure of the article is organized as follows. In Section II, some related works are revised. In Section III, the TBR-Net is introduced. In Section IV, extensive ablation and comparative experiments are performed to show the effectiveness of the proposed methods on HOTC datasets. Finally, conclusions are drawn in Section V.

II. Related Work

A. Visual Object Tracker

Object tracking has a wide range of application scenarios, such as video surveillance, robot vision, and action analysis.

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With the emergence of large-scale tracking datasets, research on this topic has reached a new height. Among them, Siamese network-based methods have better capacities to improve the tracking accuracy and efficiency [27], [28], [29], [30]. Bertinetto et al. [31] proposed a Siamese tracker named SiamFC, which is the first attempt to extract the deep feature of T & S frames through the Siamese network. With the cross-correlation layer, the tracker can match the object to search regions. After that, Li et al. [32], [33] proposed the SiamRPN family, including SiamRPN and SiamRPN++, which incorporates the RPN into the tracking framework and proposed a depth-wise separable correlation structure to enhance similarity calculation. In order to reduce the computation cost and avoid duplicate detection, many anchor-free frameworks are proposed. Guo et al. [34] proposed a simple anchor-free tracking framework with one classification–regression subnetwork for detection box prediction. The number of hyperparameters can be reduced, and excellent tracking performance can be achieved. Guo et al. [35] proposed a part-to-part framework called SiamGAT with the graph attention network and designed an object-aware area selection method to crop the ROI according to the ground truth. With the graph attention module, the prefixed global feature matching can be avoided. Chen et al. [36] proposed a network named SiamBAN to mine the representative power of the Siamese network. To make SiamBAN more flexible and general, they designed a box generation method without predefinition. However, the performance of the RGB trackers was limited because the tracker is weak in dealing with background clutter (BC) and visually undermining from objects in the same color.

B. Hyperspectral Object Tracker

With the development of HS snapshot cameras, acquiring HS images has become simple, which drives the object tracking research for HSIs. HSIs collect spatial and spectral information simultaneously, and their wide spectral coverage can help separate objects with different spectral information even if their reflectance in visible color bands is similar.

Due to the imaging mechanism of HSIs, the spatial and spectral information of the detected object is sufficient, and the spectral information is determined by the reflective properties of the object so that the problem of visual similarity can be solved effectively. Existing methods can generally be categorized into two main groups: traditional features-based methods and CNN-based methods. The former represents the object through handcrafted features. Van Nguyer et al. [37] incorporated radiative transfer theory and the mean shift algorithm to track objects in HSV. Hou et al. [15] focused on the tensor representation and 2-D scale search strategy and proposed a regularized sparse correlation filter method. The latter adopts deep features to represent objects. Uzkent et al. [38] proposed an HS tracker named DeepHKCF, which uses the features from the existing models like AlexNet and enlarges the ROI to address low temporal resolution and track aerial vehicles efficiently. Gao et al. [39] proposed a three-branch Siamese network to extract the spectral semantic information. Su et al. [40] proposed to reduce the spectral information loss by fusing the deep features corresponding to different groups of bands and initial HSIs. Li et al. [41] proposed an attention-based Siamese framework named SiamBAG, where a band attention module generates the band weights to group the HSI into several false-color images so that RGB-based tracker like SiamRPN can deal with the HSI tracking well. Gao et al. [42] proposed a feature fusion network, CBFF-Net, to enhance the representative ability of the tracker, where a bidirectional multiple deep feature fusion module is designed to fuse the cross-band information. Tang et al. [43] proposed BAHT, which aims to capture the background-aware information. The bands that can extract an object from backgrounds are selected to learn a robust HS representation. Although current CNN-based methods perform significantly better compared with manual feature-based methods, most of their band grouping methods lack consideration of local and global dependencies between bands, and, in addition, they simply reuse models in the existing RGB domain but neglect the modeling of interrelationships between template frames and search frames.

C. Transformer in Tracker

The Transformer is based on self-attention mechanisms, which learn dependencies of all input sequences and aggregate information from a global perspective. Recently, it has been used to deal with the object tracking task. On the one hand, some methods utilize the Transformer to calculate the relationship between T & S frames. Thanks to the mutual relationship mining ability of the Transformer, the similar regions between T & S frames can be better captured. Chen et al. [20] proposed an attention-based feature fusion method. It combines the T & S region features using self-attention and cross-attention. Then Chen et al. extended TransT [20] called TransT-M [44]. TransT-M captured the feature context and equipped with a feature augment method, which establishes associations between distant features. On the other hand, most methods adopt the Transformer as the backbone network for feature extraction, and the self-attention mechanism can adaptively adjust the relationship between T & S frames during the feature extraction, avoiding the additional relationship operation of the above methods. Ye et al. [45] proposed a one-stream Transformer-based tracking method (OSTrack). OSTrack constructed the image pairs of template-search images, which are highly parallelized and can capture bidirectional information. Lin et al. [46] proposed a Siamese tracking network leveraging the Transformer named SwinTrack. Specifically, SwinTrack adopted the Swin Transformer [47] as the backbone network. A novel motion token of the encoder and decoder embeds historical object trajectory to improve the tracking performance. Wang et al. [21] separated the Transformer encoder and decoder into two parallel branches. In this way, the object template generation and the tracking cues propagation from the previous frames can be conducted simultaneously. Ma et al. [48] proposed a unified Transformer tracker for both single and multiple object tracking. In the HSI tracking field, there is little research to utilize the Transformer for object tracking. To leverage context information mining ability and

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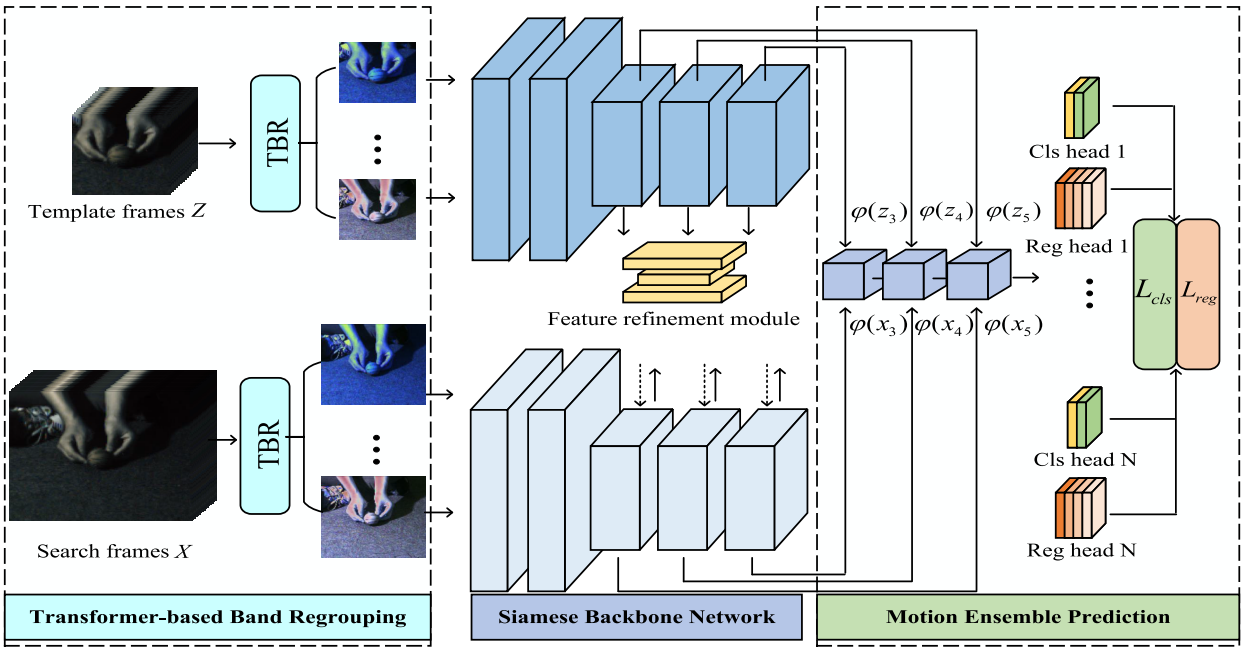

Fig. 2. Illustration of the proposed TBR-Net, including the TBR method, Siamese backbone network, and motion ensemble prediction.

obtain the coefficient matrix to better describe band importance, the TBR method is proposed in this article. It utilizes the Transformer encoder to mine the cross-band information with a simple spectral embedding. The regrouped bands can be used to optimize the whole network in end-to-end training.

III. Proposed Method

A. Framework Overview

Fig. 2 illustrates the proposed network framework. Firstly, the TBR method generates the false-color images. Secondly, the backbone network adopts the conventional two-stream Siamese ResNet-50 structure, and the generated false-color images after TBR regrouping are utilized as input T & S frames, which follow the RGB single-object tracking paradigm. Finally, the correspondence between T & S frames is computed in chunks, thus filtering out the regions with high similarity and tracking the same object as the template frame. In addition, the backbone network is pretrained on main-stream RGB datasets [36], which has two advantages: it can directly leverage RGB domain knowledge in a transfer learning manner, and the generation of false-color images can be considered a form of data augmentation to effectively compensate the scarcity of HS training data.

The T & S frames are first cropped to the resolution of 127 × 127 and 255 × 255, respectively, to highlight the object area. Let $Z \in \mathbb{R}^{H_z \times W_z \times B}$ denote input template images and $X \in \mathbb{R}^{H_x \times W_x \times B}$ denote input search images. H_z and H_x denote the heights of T & S frames, and W_z and W_x denote the widths of T & S frames, respectively. B denotes the number of bands. The two cropped images are input to the parameters-shared Siamese network. The outputs of each stage of ResNet-50 are denoted as z_i and x_i, $i = 1, 2, 3, 4, 5$. Then, to make the network more focused on the parts that are similar between T & S frames, FRM is proposed to model the pixel-level similarity in the form of self-attention between the two input frames within the latter three stages. The refined features $\varphi(z)$ and $\varphi(x)$ are input in the classification and regression training as [36]. The classification and regression branches perform depth-wise relationship calculations on input features of T & S frames. Then, the classification head outputs two channels to predict whether a point belongs to the background using a convolutional layer followed by batch normalization and activation layers. The regression head outputs four channels to predict the position of objects in the same way. The TBR-Net can accurately predict the object position in subsequent frames by ensembling multistage prediction from different false-color images. Finally, to learn the robust feature representation, multiobjective loss is adopted to train the whole network.

B. Transformer-Based Band Regrouping

To utilize the pretrained model parameters in the RGB domain, the original HSI should be converted into multiple three-channel false-color images. Given that different bands contain varying amounts of information, a metric is needed to measure the information content within the bands as a basis for band reranking. In the band selection domain, the network can learn a coefficient matrix that expresses the importance between bands using band reconstruction. The coefficient matrix reflects the intrinsic relationship between bands. However, this method has two significant shortcomings. First, either autoencoder-based SSC-like methods or traditional attention mechanisms lack global semantic information mining, or they require specialized loss functions for training. Multiobjective optimization strategies may not necessarily be conducive to finding the globally optimal solution. Second, SSC methods

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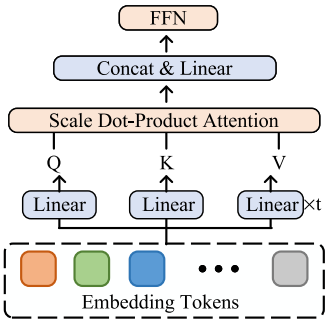

Fig. 3. Structure of the MSA. The output of the MSA is fed into the FFN.

predominantly utilize shallow texture features and may fail to capture high-level semantic information. To address these issues, the TBR is proposed, where the Transformer is employed to implicitly model the relationship between bands and integrate the training of band regrouping into the tracking training. This method can leverage the strengths of the Transformer architecture, effectively capturing both texture features and high-level semantic information, thereby improving the overall results of the proposed tracker.

1) Transformer Revisit: The Transformer encoder is composed of two core components: the multihead self-attention (MSA) mechanism, which allows the model the significance of various samples of the input data, and the feed-forward network (FFN), which applies position-wise transformations to the processed data. The former includes several scaled dot-product attention, which is formulated by

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \tag{1}$$

where Q, K, and V denote the embedding vectors transformed from the input tensor linearly, and d denotes the embedding dimensions. The output can be considered as the weighted sum of V under the guidance of correlation between Q and K. To capture the semantic information of different locations in different subspaces, MSA obtains a broader global receptive field through multiple self-attention computations. This can be understood as performing multiple parallel self-attention head calculations, which are formulated by

$$\text{MSA}(Q, K, V) = \text{Concat}(h_1, \ldots, h_t)W^o \tag{2}$$

$$h_t = \text{Attention}\left(QW_t^q, KW_t^k, VW_t^v\right) \tag{3}$$

where W_t^q, W_t^k, W_t^v, and W^o represent the projection matrices. t is the number of head calculations. Concat is the concatenation operation. The structure of MSA is shown in Fig. 3. To process and transform the information encoded in the self-attention mechanism, a nonlinear layer FFN is introduced, which is applied independently to each position in the sequence. It is composed of a pair of linear transformation layers with a nonlinear GeLU function, which is defined as follows:

$$\text{FFN}(\text{MSA}_{\text{out}}) = \text{FC}_2(\text{GeLU}(\text{FC}_1(\text{MSA}_{\text{out}}))) \tag{4}$$

where MSA_{out} represents the output of MSA. FC_1 and FC_2 represent two fully connected layers with different weights. GeLU is a Gaussian error linear unit. The MSA captures the linear relationships within input sequences, and the output of the FFN is a new hidden representation after nonlinear transformation and mapping, which can enhance the feature expression capability and increase the independence between positions.

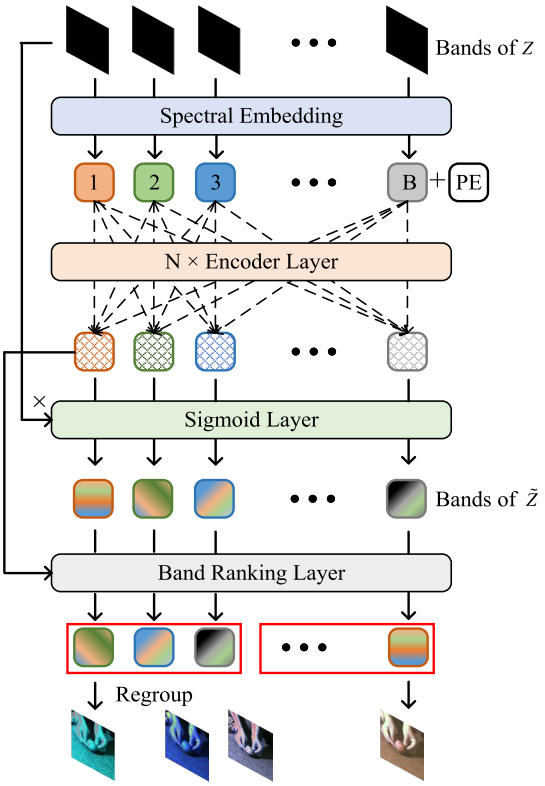

Fig. 4. Structure of the TBR network, which includes the spectral embedding, Transformer encoder layers, and band ranking and grouping layer. Here the template image Z is used as an example.

2) Details of TBR: Considering the Transformer's ability to model long-term dependencies and the fact that MSA and FFN can integrate multiple subspace nonlinear feature representations and fit complex spectral data, the TBR is proposed. This is intended to leverage the Transformer's inherent capability to implicitly model relationships between individual bands. The coefficient matrix can be obtained from the Transformer encoder. This method has two benefits: first, TBR inherently encapsulates global nonlinear spectral relationships, and the resultant spectral coefficients can represent the importance of the bands; second, utilizing these spectral coefficients as attention weights enhances the quality of feature representation. The output weights of TBR are directly used as inputs to the tracker after the bands have been reorganized with the original image, obviating the need for trade-off training.

Fig. 4 depicts the structure of TBR. To meet the requirements of constructing the embedding token needed by the Transformer, the global average pooling (GAP) is utilized as the spectral embedding to integrate each spectral information without introducing additional computational effort. Moreover, the position information is crucial to the Transformer. As such, a learnable parameter of the same size is added as the position embedding to the spectral embedding. This allows the embedding token, with positional information and embedded spectral

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information, to be fed into the encoder of the Transformer. The process is presented as follows:

$$I_z = \mathrm{SE}(Z) + \mathrm{PE}$$
$$I_x = \mathrm{SE}(X) + \mathrm{PE} \tag{5}$$

where I_z and I_x denote the embedding token of Z and X, respectively. SE denotes the spectral embedding, and PE denotes the position encoding parameters. The dimensions of I_z and I_x are both $B \times 1$. Followed by a Sigmoid function, I_z and I_x can be input in the Transformer encoder. The intrinsic information $\tilde{I}_z$ and $\tilde{I}_x$ of bands is obtained through the attention mechanism and nonlinear projection of FFN as follows:

$$\tilde{I}_z = \mathrm{FFN}(\mathrm{MSA}(I_z W^q, I_z W^k, I_z W^v))$$
$$\tilde{I}_x = \mathrm{FFN}(\mathrm{MSA}(I_x W^q, I_x W^k, I_x W^v)) \tag{6}$$

where W^q, W^k, and W^v denote the linear transformation matrices which generate the Q, K, and V in (2). In this way, the coefficient maps M_z and M_x are obtained by

$$M_z = \mathrm{Sigmoid}(\tilde{I}_z)$$
$$M_x = \mathrm{Sigmoid}(\tilde{I}_x). \tag{7}$$

The larger values in M_z and M_x mean the more importance. The maps M_z and M_x are adopted as masks to filter the original spectral information, incorporating a residual connection to preserve discriminative information and stabilize the training process. It can be defined as follows:

$$\tilde{Z} = Z + M_z \odot Z$$
$$\tilde{X} = X + M_x \odot X \tag{8}$$

where $\odot$ is the Hadamard product. Afterward, the feature maps $\tilde{Z}$ and $\tilde{X}$ with discriminative information are regrouped into multiple false-color images according to the order of the coefficient masks M_z and M_x. Specifically, we use the most important 15 of an HS containing 16 bands. For example, the band importance with the descending order is (2, 16, 12, 9, 5, 1, 3, 11, 4, 6, 13, 15, 7, 8, 10, 14), and the three bands in order of importance from top to bottom of the bands are selected to form the images group: group1 (2, 16, 12), group2 (9, 5, 1), group3 (3, 11, 4), group4 (6, 13, 15), and group5 (7, 8, 10).

By adopting this strategy, the five three-channel false-color images are generated from an HSI to form the input T & S frames, which are then fed into the backbone network for tracker training. The original HS information is fused with the intrinsic spectral information produced by TBR, and the band importance is effectively derived from a global perspective. This approach allows us to comprehensively leverage the sufficient spectral information inherent in HSVs, enhancing the performance of the tracking system.

C. Feature Refinement Module

In this section, the structure of FRM is introduced. For the existing models [19], [36], they employ a pretrained model within the RGB domain for the extraction of false-color image features. However, the subsequent depth-wise cross-correlation computations neglect the semantic similarity between the objects and lack heuristic information. Moreover, while most relation modeling methods introduce new prediction heads to resolve the accuracy issue inherent in relational computations, few of them leverage the contextual modeling capabilities of the feature extraction network. This underscores the need for a more effective approach that can address these limitations and fully exploit the potential of feature extraction networks.

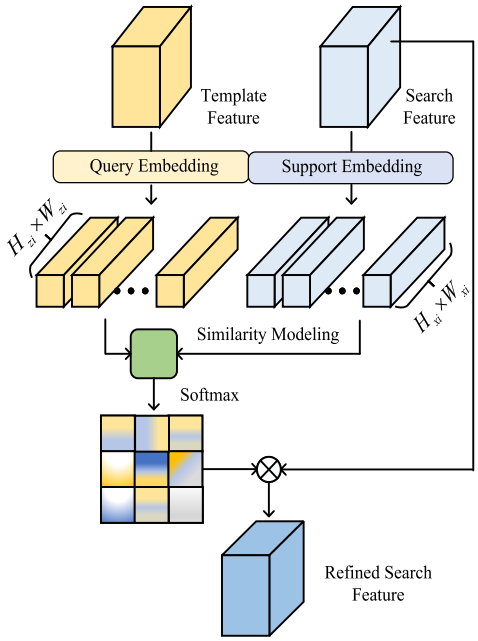

Fig. 5. Structure of the FRM. The module takes the T & S feature as input and generates the refined search feature.

Inspired by the structure of the Transformer, the FRM is proposed to utilize the self-attention mechanism at the pixel level to make the network focus on similar regions between T & S frames. Specifically, we focus on the last three feature extraction stages output $z_i, x_i, i \in \{3, 4, 5\}$, and model the pixel-level correlation representation between T & S features. The structure of FRM is depicted in Fig. 5. First, a 1×1 convolutional layer is used as a linear mapping performed on the frame feature. Subsequently, the feature vector is obtained from the flattened projected feature. The process is explicitly represented as follows:

$$\hat{z}_i = \mathrm{Flatten}(\mathrm{conv}(z_i))$$
$$\hat{x}_i = \mathrm{Flatten}(\mathrm{conv}(x_i)) \tag{9}$$

where conv represents 1×1 convolution. $\hat{z}_i$ and $\hat{x}_i$ denote the mapped feature vectors. In this way, the pixel-level cosine similarity can be calculated by multiplying the feature vectors. The pixel-level attention map M_i^{pixel} is obtained through the Softmax activation function

$$M_i^{\mathrm{pixel}} = \mathrm{Softmax}(\hat{z}_i \otimes \hat{x}_i) \tag{10}$$

where $\otimes$ denotes matrix multiplication. The value of M_i^{pixel} reveals how much attention each pixel from x_i requires. The features can aggregate heuristic information in the global context guided by the attention map. The refined features can

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be obtained as a product of the search features and the attention weights as

$$\bar{x}_i = M_i^{\text{pixel}} \odot x_i + x_i \tag{11}$$

where $\bar{x}_i$ denotes the refined feature of x_i and $\odot$ denotes the element-wise multiplication. Here, the residual connection is added to preserve the original information. Through a simple attention operation, the network can capture contextual information similar to T & S features and focus on the object region during network training to improve tracking accuracy. With the refined search features, the classification and regression map can be effectively obtained from the depth-wise cross-correlation calculation. Using a straightforward attention operation, the network can capture contextual information between T & S features and concentrate on the object region during network training to enhance tracking accuracy.

D. Tracking Strategy and Loss Function

1) Tracking Strategy: During the inference stage, T & S frames are initially cropped and regrouped into N ($N = 5$ here) false-color images, respectively. It is worth noting that we take one of N false-color images t^1 and s^1 as an example to show the following process for simplicity. The chosen T & S frames t^1 and s^1 are then inputted to the T & S branch of the backbone network and FRM to extract the features $\varphi(t^1)$ and $\varphi(s^1)$. This process allows us to derive the features needed for subsequent matching tasks. Given $\varphi(t^1)$ and $\varphi(s^1)$, the classification and regression head is employed to deduce the motion of the objects. Specifically, $\varphi(t^1)$ can be viewed as a convolution kernel that perceives the input area of $\varphi(s^1)$ to predict the possibility of the background and the position of objects. This process can be described as follows:

$$\begin{aligned} P_{w\times h\times 2}^{\text{cls}} &= [\varphi(t^1)]_{\text{cls}} * [\varphi(s^1)]_{\text{cls}} \\ P_{w\times h\times 4}^{\text{reg}} &= [\varphi(t^1)]_{\text{reg}} * [\varphi(s^1)]_{\text{reg}} \end{aligned} \tag{12}$$

where $[\varphi(t^1)]_{\text{cls}}$, $[\varphi(s^1)]_{\text{cls}}$, and $[\varphi(t^1)]_{\text{reg}}$, $[\varphi(s^1)]_{\text{reg}}$ denote the two head embeddings of $\varphi(t^1)$ and $\varphi(s^1)$ through the convolution layer, $*$ denotes the depth-wise cross-correlation calculation, $P_{w\times h\times 2}^{\text{cls}}$ represents the classification map, and $P_{w\times h\times 4}^{\text{reg}}$ represents the regression map. Note that fusing the feature maps from different layers can capture the low-level fine-grained and high-level semantic information simultaneously, and we ensemble multiple prediction maps from the last three feature extraction stages to obtain robust tracking performance. It can be formulated as follows:

$$\begin{aligned} P_{\text{cls-mul}}^1 &= \sum_{i=3}^{5} \mu_i \left[\varphi\left(t_i^1\right)\right]_{\text{cls}} * \left[\varphi\left(s_i^1\right)\right]_{\text{cls}} \\ P_{\text{reg-mul}}^1 &= \sum_{i=3}^{5} \mu_i \left[\varphi\left(t_i^1\right)\right]_{\text{reg}} * \left[\varphi\left(s_i^1\right)\right]_{\text{reg}} \end{aligned} \tag{13}$$

where i denotes the stage of the backbone. λ_i and μ_i are learnable parameters to adaptively adjust the contributions of multiple layers. $P_{\text{cls-mul}}^1$ and $P_{\text{reg-mul}}^1$ denote the multiple classification and regression prediction maps, respectively. Furthermore, the final regression prediction map P can be obtained by the weighted fusion of regression maps from multiple false-color images, which is expressed as follows:

$$P = \sum_{n=1}^{N} M_n P_{\text{reg-mul}}^n \tag{14}$$

where M_n represents the band weights generated by TBR.

2) Loss Function: The overall loss function is defined as the following multitask loss function:

$$L = \alpha L_{\text{cls}} + \beta L_{\text{reg}} \tag{15}$$

where L_{reg} is the intersection over union (IoU) loss [49], L_{cls} denotes the cross-entropy loss, and α and β are trade-off parameters. Following [36], let g_w, g_h, (g_{x_1}, g_{y_1}), (g_{x_2}, g_{y_2}), and (g_{x_c}, g_{y_c}) be the width, height, top-left corner, bottom-right corner, and center point of ground-truth bounding box, respectively. The two different ellipses E_1 and E_2 are defined as follows:

$$\frac{\left(p_i - g_{x_c}\right)^2}{\left(\frac{g_w}{2}\right)^2} + \frac{\left(p_j - g_{y_c}\right)^2}{\left(\frac{g_h}{2}\right)^2} = 1 \tag{16}$$

$$\frac{\left(p_i - g_{x_c}\right)^2}{\left(\frac{g_w}{4}\right)^2} + \frac{\left(p_j - g_{y_c}\right)^2}{\left(\frac{g_h}{4}\right)^2} = 1 \tag{17}$$

where (p_i, p_j) denotes the locations of pixel points. In this way, two ellipses E_1 and E_2 of different sizes about the ground-truth bounding box are constructed [36]. The loss function regards the pixel point in E_1 as the positive samples and regards the pixel point out of E_2 as the negative samples. Then the positive pixel point (p_i, p_j) is adopted to predict the position of objects, which is formulated by

$$\begin{aligned} d_l &= p_i - g_{x_1} \\ d_t &= p_j - g_{y_1} \\ d_r &= g_{x_2} - p_i \\ d_b &= g_{y_2} - p_j \end{aligned} \tag{18}$$

where d_l, d_t, d_r, and d_b are the vertical distances between the pixel point and the four sides of the position labels. With the regression object, IoU is used to quantify the extent of overlap between the predicted position and position labels. The specific form of L_{reg} is given by

$$L_{\text{reg}} = 1 - \text{IoU}. \tag{19}$$

During the training process, the gap between the predicted results and position labels narrows gradually to regress d_l, d_t, d_r, and d_b uniformly.

E. Analysis on the Proposed Method

In this article, we propose a TBR and feature refinement network to tackle the challenge of HS object tracking. First, the proposed TBR method models the long-range dependencies between bands and generates the coefficient matrix representing the importance of bands. Different from [19], the attention mechanism of the Transformer looks at an input band sequence and decides at each step which other parts of the band sequence are important, and therefore facilitates capturing the global information from the input band sequence. This weight

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matrix reflects the importance of each false-color image for correct tracking, and the false-color images combined in order of weights can maximize their effectiveness. Its effectiveness can be explained by comparing the attention maps of the experimental section with the existing band recombination methods. In this way, the TBR is optimized through end-to-end training with tracking performance as a traction metric, which alleviates the problem of insufficient training data to some extent. Second, the dual ResNet-50 [50] trained on the RGB tracking dataset is adopted as the Siamese backbone, which can transfer the domain knowledge from the RGB domain to the HS false-color domain. Finally, the relationship modeling between the two input frames is neglected when the pretrained RGB tracker is used to fine-tune the HSI domain. Inspired by the cross-attention and correlation filter mechanism [51], FRM is proposed. The pixel-level similarity between T & S frames is considered, and the module is trained end-to-end. In this way, the TBR-Net finds similar regions of objects and stores the knowledge from subsequent frames.

TABLE I
ELEVEN ATTRIBUTES OF THE HOTC DATASET

Attributes	Scenes
background clutters (BC)	bus,card,coke,drive,forest,forest2,fruit,hand, kangaroo,toy1,toy,worker
deformation (DEF)	book,excavator,hand,kangaroo,player,rubik, pedestrian2
fast motion (FM)	basketball,bus,bus2,coke
illustration variation (IV)	bus2,campus,car3,pedestrian,rider1,rider2, student,toy2,pedestrian2,trucker
in-plane rotation (IPR)	board,book,car,car2,card,coke,drive,excavator, face,face2,paper,player,rubik
low resolution (LR)	basketball,bus,car3,rider1,rider2,worker, pedestrian2
motion blur (MB)	ball,basketball,face,kangaroo
occlusion (OCC)	ball,basketball,board,bus2,campus,car,car3,card, excavator,face2,forest,forest2, fruit,playgrou-nd,rider1,rider2,toy1,toy2,pedestrian2,trucker
out-of-plane clutters (OPR)	board,book,car,car2,coke,drive,excavator,face, face2,hand,kangaroo,player,rubik,toy2
out-of-view (OV)	trucker
scale variation (SV)	ball,board,bus2,campus,car,car2,car3,coke,drive, excavator,face,face2,hand,kangaroo,pedestrian, player,playground,rider1,rider2,student,toy2, trucker,worker

IV. EXPERIMENTS

In this section, ablation and comparison experiments on the public dataset HOTC are performed to demonstrate the effectiveness of the proposed methods.

A. Experiment Settings

1) Datasets: The hyperspectral object tracking competition (HOTC) [1] provided the HSV datasets on which we train and test this method. The training set contains 40 video clips, and the testing set contains 35 video clips. Both of the above include HSVs (16 bands), false-color videos (three channels), and RGB videos (three channels). All types of videos are acquired from the same point of view. Besides, there are 11 different challenging factors, including BC, deformation (DEF), fast motion (FM), illumination variation (IV), in-plane rotation (IPR), low resolution (LR), motion blur (MB), out-of-occlusion (OCC), plane rotation (OPR), out-of-view (OV), and scale variation (SV). The detailed settings are listed in Table I.

2) Evaluation Metrics: The success plot of the area under the curve (AUC) and the precision plot of the precision rate at multiple thresholds are adopted as the evaluation metrics. The former means the successful frame rate between the regression results and position labels at different thresholds. The overlap ratio can be represented by IoU. It can be formulated as follows:

$$\mathrm{IoU} = \frac{\left|B_p \cap B_t\right|}{\left|B_p \cup B_t\right|} \tag{20}$$

where B_p denotes the predicted results, B_t denotes the position labels, and $\cap$ and $\cup$ represent the intersection and the union, respectively. With the definition of IoU, the success plot quantifies the proportion of tracking results that exceed a threshold, which varies between 0 and 1.

The latter evaluates the center location error against various pixel thresholds, which typically range from 1 to 50 pixels. The distance between the centers of B_p and B_t is defined as follows:

$$\mathrm{DIS} = \sqrt{(c_{xp} - c_{xt})^2 + (c_{yp} - c_{yt})^2} \tag{21}$$

where (c_{xp}, c_{yp}) denotes the predicted center points and (c_{xt}, c_{yt}) denotes the center point of the position labels. With the definition of DIS, the precision plot measures the ratio of tracking results with distances of center pixels between B_p and B_t less than a threshold. The distance threshold ranges from 0 to 50. Typically, the precision plot with the threshold of 20 is used to rank the trackers, denoted as DP_20.

3) Implementation Details: We use an NVIDIA RTX 4090 graphics processing unit (GPU) and PyTorch framework to perform the training and testing experiments. For the model and parameters of the backbone network, we refer to SiamBAN, which is a pretrained model on the RGB dataset including COCO [52], YouYube-BB [53], ImageNet DET [54], and ImageNet VID [54]. In this way, the knowledge in the RGB domain can be transferred to the HS domain with the fine-tuning of false-color images from the HS domain. In the training phase, the parameters of the backbone network are frozen, including ResNet-50, the adjust, and the head network, to train TBR and FRM. The stochastic gradient descent (SGD) optimizer is used to train the network for 30 epochs. The batch size is set to 12. α and β are set to 0.5. The learning rate is initially set to 0.0003, and a warm-up strategy is used to adjust the learning rate. The learning rate first increased uniformly to 0.0003 in the first five epochs and then decreased uniformly to 3e-6.

B. Comparison With State-of-the-Arts

In this section, several experiments, including comparisons with RGB and HS trackers, attribute-based comparison, and visualization analysis, are conducted.

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TABLE II

OVERALL PERFORMANCE (AUC AND DP_20) COMPARISON OF HS AND VISUAL TRACKER. THE HS TRACKERS DEAL WITH THE HSIS AND THE VISUAL TRACKER RUNS ON THE FALSE-COLOR IMAGES. WE USE THE RED AND BLUE FONTS TO HIGHLIGHT THE TOP TWO VALUES

	Hyperspectral tracker					Visual tracker			
Method	TBR-Net	BAE-Net [16]	SEE-Net [19]	SiamBAG [41]	SST-Net [55]	TransT [20]	SiamGAT [35]	SimTrack [56]	OSTrack [45]
AUC	0.660	0.606	0.654	0.622	0.623	0.633	0.581	0.600	0.557
DP_20	0.920	0.878	0.907	0.877	0.916	0.879	0.827	0.845	0.816

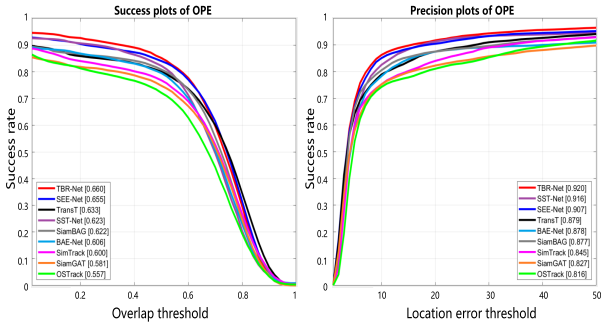

Fig. 6. Comparison with advanced trackers on HOTC dataset. (a) Depicts the success plot. (b) Depicts the precision plot.

1) Quantitative Comparison With HS Trackers: We first compare TBR-Net with state-of-the-art HS trackers, including BAE-Net, SST-Net, SiamBAG, and SEE-Net. BAE-Net adopted a channel attention module to generate a band importance matrix to regroup the false-color images. SST-Net [55] proposed a spectral–spatial–temporal attention network to focus on the salient spatial features and temporal relationships among adjacent frames. SiamBAG proposed a band grouping Siamese network to transfer the knowledge of RGB trackers to enhance the performance of HS trackers. SEE-Net proposed a deep ensemble network with the band regrouping under the guidance of SSC. It is worth noting that we reproduce SiamBAG and SEE-Net while the results of other trackers are currently available.

The quantitative results are shown in Table II. Although all the methods we compare are based on band regrouping, BAE-Net achieves poor accuracy due to the lack of global semantic information captured by its channel attention mechanism. SiamBAG considers the histogram equalization of images and relocates the fine-tuned object position, so it yields 0.622 of AUC score and 0.877 of DP_20. SST-Net can learn a better band relationship by combining the spectral, spatial, and temporal attention information, so the experimental results are improved further to 0.623 of AUC score and 0.916 of DP_20. SEE-Net yields the AUC score of 0.654 and DP_20 of 0.907 with the prior knowledge provided by BAE-Net, incorporating the channel attention to SSC. The proposed TBR-Net can achieve better performance, yielding an AUC score of 0.660 and DP_20 of 0.920. It demonstrates the effectiveness of the TBR-Net with the intrinsic relationship modeling from a global perspective, and the relationships between patches from the template and regions within the search area are considered.

2) Quantitative Comparison With RGB Trackers: We compare the proposed methods with state-of-the-art RGB trackers, including SiamGAT, OSTrack, SimTrack, and TransT. SiamGAT adopted the graph attention network to SiamCAR to improve the tracking performance further. OSTrack proposed a one-stream framework with the pure Transformer network and designed an early candidate elimination module to improve inference efficiency. SimTrack [56] proposed a simplified tracking architecture, leveraging the Transformer backbone with the foveal window strategy to expand the image patches. TransT proposed a feature fusion layer with multiple attention modules. It is worth noting that the visual tracker experiments are performed on false-color datasets.

From Table II and Fig. 6, it can be seen that the OSTrack provides inferior tracking performance with an AUC score of 0.557 and DP_20 of 0.816. This is because the effect of the candidate early elimination may be disturbed by the BC. SimTrack considers the diversity of image patches, yielding the AUC score of 0.600 and DP_20 of 0.845. The SiamGAT model trained on the TrackingNet dataset is utilized. Considering the object area selection, the performance of SiamGAT improves up to 0.581 of the AUC score and 0.827 of DP_20. However, the SV of objects may limit the tracking performance. TransT further improves the tracking performance with the Transformer-like fusion module, achieving the AUC score of 0.633 and DP_20 of 0.879. The proposed TBR-Net achieves the best performance, thanks to TBR and FRM. Moreover, the experimental results demonstrate that the RGB trackers are weak in dealing with the HOTC dataset's BC.

TABLE III

OVERALL PERFORMANCE (AUC AND DP_20) COMPARISON OF HS TRACKER ON THE HOTC-2023 DATASET. WE USE THE RED AND BLUE FONTS TO HIGHLIGHT THE TOP TWO VALUES

Methods	TBR-Net	BAE-Net [16]	SiamBAG [41]	SEE-Net [19]	SiamCAT [57]
AUC	0.599	0.558	0.554	0.588	0.593
DP_20	0.826	0.802	0.779	0.808	0.822

3) Additional Comparison on the HOTC-2023 Dataset: To further demonstrate the effectiveness of our method, we conducted experiments on the 16 bands' dataset of HOTC-2023 with the HS tracker BAE-Net, SiamBAG, and SEE-Net. It has 55 HSVs for training and 46 HSVs for testing. We add the SiamCAT [57], which proposes a guided learning attention (GLA) module to learn spectral features. The experimental results are listed in Table III. Our TBR-Net still outperforms the state-of-the-art HS trackers, which demonstrates its effectiveness.

4) Running Costs Comparison: We report the running cost comparison results, including the computational complexity [floating point operations per second (FLOPs)], model size, and running speed [frame per second (FPS)]. The results

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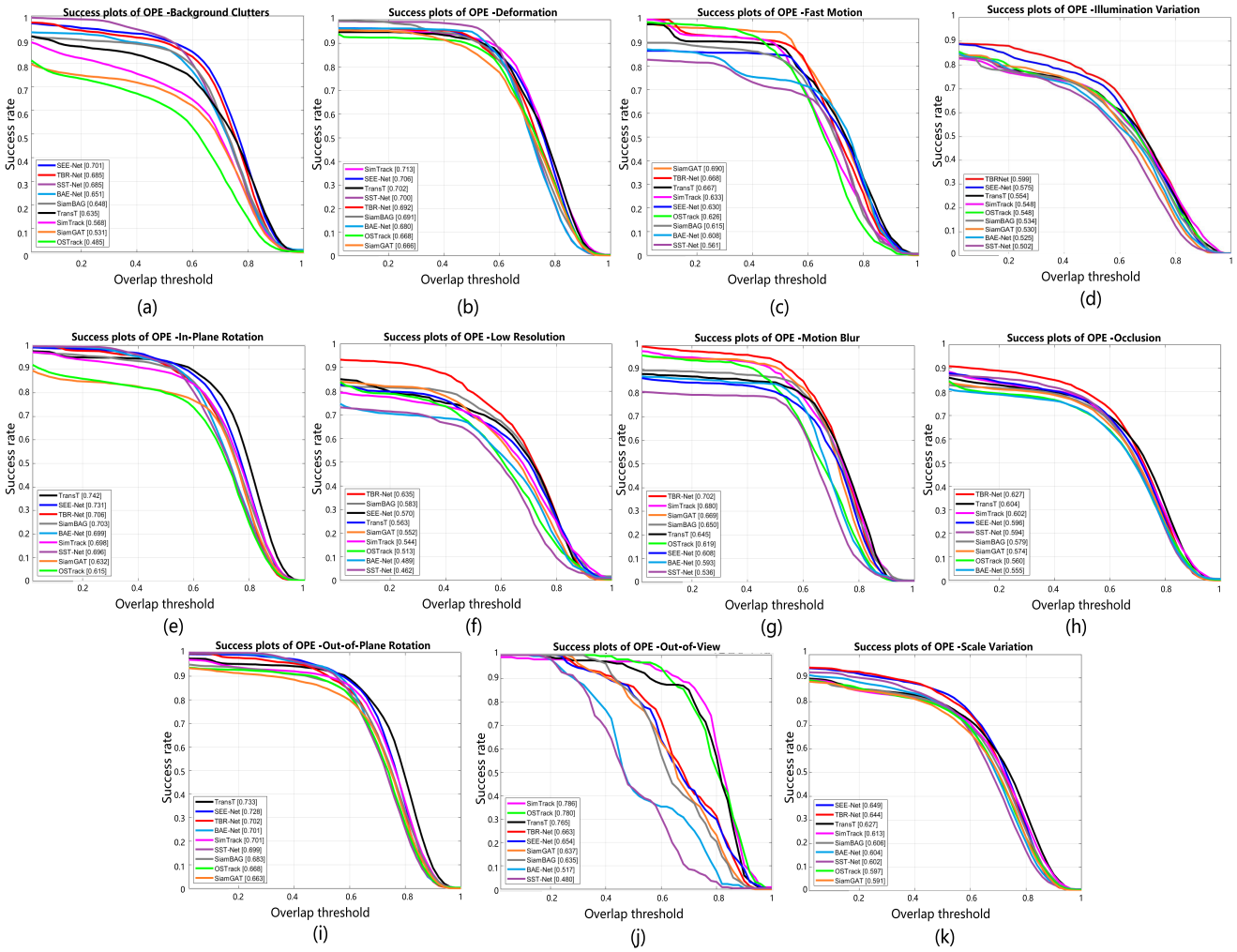

Fig. 7. Attribute-based comparison of success plots. (a) Success plot of BC. (b) Success plot of DEF. (c) Success plot of FM. (d) Success plot of IV. (e) Success plot of IPR. (f) Success plot of LR. (g) Success plot of MB. (h) Success plot of OCC. (i) Success plot of OPR. (j) Success plot of OV. (k) Success plot of SV.

TABLE IV
RUNNING COSTS COMPARISON RESULTS INCLUDING FLOPS, MODEL SIZE, AND FPS

Methods	TBR-Net	SEE-Net [19]	SiamBAG [41]	TransT [20]	OSTrack [45]
FLOPs	298.956G	297.992G	109.277G	19.194G	26.627G
Size	54.054M	53.938M	90.442M	18.542M	92.121M
FPS	14.92	18.83	26.71	66.73	24.54

are shown in Table IV. All the trackers ran on a Windows machine with the configuration of an Intel[1] Core[2] i7-13700k CPU@3.40 GHz and an NVIDIA 4090 GPU. It can be observed that due to the need to calculate the forward propagation process of multiple false-color images using band regrouping methods (TBR-Net, SEE-Net, and SiamBAG), the FLOPs are higher compared to visual trackers (TransT and OSTrack) that directly infer a single false-color image. But compared to the baseline SEE-Net, our proposed TBR-Net achieves performance improvement with a small negligible increase in FLOPS, demonstrating the effectiveness of our method in improving tracking performance. In addition, the size of the model has significant advantages compared to the existing methods.

5) Attribute-Based Comparison: The 11 attribute-based comparison experiments are conducted with all trackers, including four HS trackers and four RGB trackers. The experimental results are shown in Fig. 7 and Table V. It can be seen that TBR-Net yields the best two performances among the seven attributes. TBR-Net performs well in both BC and IV factors, where the HS trackers all outperform the RGB trackers in the BC factor, with a maximum difference in performance of 0.216 (between SEE-Net and OSTrack). This indicates that the HS tracker takes into account the spectral differences between the object and the background, reflecting the advantages of HSIs for dealing with object and background similarities. The case with MB and LR factors represents the object image with low visual quality, which undermines the performance of the RGB tracker relying on visual features. Meanwhile, because of the band regrouping method as well as the ensemble learning method, which combines the prediction results of multiple weak trackers, it is robust for dealing with low-quality images and still has superiority in comparison with the HS tracker. For FM, the RGB trackers generally outperform the HS trackers. We believe that color provides richer visual information and

[1]Registered trademark.
[2]Trademarked.

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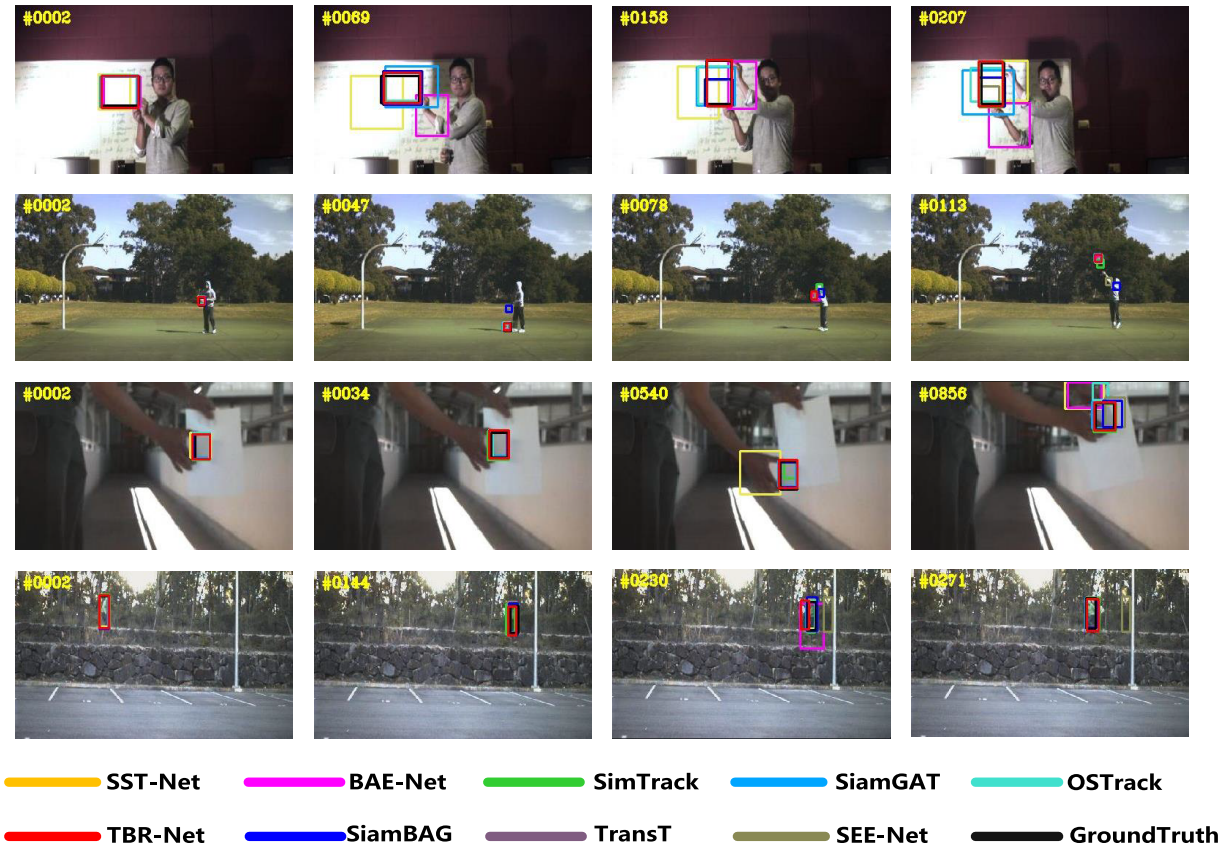

Fig. 8. Visualization of tracking results of four challenging scenarios on the false-color images, including paper, basketball, card, and forest2.

TABLE V

ATTRIBUTE-BASED EXPERIMENTAL RESULTS (AUC). WE USE THE RED AND BLUE FONTS TO HIGHLIGHT THE TOP TWO VALUES

Methods	BC	DEF	FM	IV	IPR	LR	MB	OCC	OPR	OV	SV
OSTrack [45]	0.485	0.668	0.626	0.548	0.615	0.513	0.619	0.560	0.668	0.780	0.597
SimTrack [56]	0.568	0.713	0.633	0.548	0.698	0.544	0.680	0.602	0.701	0.786	0.613
SiamGAT [35]	0.531	0.666	0.690	0.530	0.632	0.552	0.669	0.574	0.663	0.637	0.591
TransT [20]	0.635	0.702	0.667	0.554	0.742	0.563	0.645	0.604	0.733	0.765	0.627
BAE-Net [16]	0.651	0.680	0.608	0.525	0.699	0.489	0.593	0.555	0.701	0.517	0.604
SST-Net [55]	0.685	0.700	0.561	0.502	0.696	0.462	0.536	0.594	0.699	0.480	0.602
SiamBAG [41]	0.648	0.691	0.615	0.534	0.703	0.583	0.650	0.579	0.683	0.635	0.606
SEE-Net [19]	0.701	0.706	0.630	0.575	0.731	0.570	0.608	0.596	0.728	0.654	0.649
TBR-Net (ours)	0.685	0.692	0.668	0.599	0.706	0.635	0.702	0.627	0.702	0.663	0.644

is more sensitive to changes in the surroundings. However, TBR-Net can achieve the second-highest accuracy, reflecting the richer semantic information constructed by the proposed band selection method. In the deformation, occlusion, and SV scenarios, since HSI is regrouped into multiple three-channel false-color images, it is possible to provide multiple pairs of matching samples for integrated prediction. It is robust in dealing with occlusions, scale changes, and deformations of the object. For the in-plane rotation and out-of-plane rotation factors, due to the object's rotation, e.g., rolling, pitching, and yawing, the appearance of the frames varies too much, which makes it difficult for FRM to capture the relationships between frames. Thus, the performance of the proposed method is not good. In the case of OV (trucker class in the dataset), the RGB tracker is superior to the HS tracker. The main reason is that the severe occlusion of trucks and small objects prevents the advantages of HS from being realized. However, even in this case, our method still outperforms all the HS trackers, demonstrating its effectiveness.

6) Qualitative Comparison: To show the performance and explore the internal mechanisms of TBR-Net, we conduct several visualization experiments in this section.

a) Tracking results: To illustrate the effectiveness of our methods, the four challenging scenarios are selected, including *paper* (IPR, BC), *basketball* (FM, MB, OCC, LR), *card* (IPR, BC, OCC), and *forest2* (BC, OCC) to show the tracking results of the above RGB and HS trackers. Most of the objects in these scenes are similar in appearance to the background. The label box is framed by black lines, and the predicted box from our TBR-Net is framed by red lines. As shown in Fig. 8, the tracker predicts the state of the object in the subsequent frames based on the object's position in the first frame given by the label. The visualized image of the second frame is shown in the first column. For the last three columns, we can observe that in the

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TABLE VI
ABLATION EXPERIMENTAL RESULTS (AUC AND DP_20) WITH BASELINE MODEL ON THE HOTC AND HOTC-2023 DATASET. Δ INDICATES THE AMOUNT OF VARIATION

	HOTC				HOTC-2023			
Method	AUC	DP_20	Δ(AUC)	Δ(DP_20)	AUC	DP_20	Δ(AUC)	Δ(DP_20)
B	0.654	0.907	-	-	0.588	0.808	-	-
B+T	0.655	0.918	+0.001	+0.011	0.594	0.823	+0.006	+0.015
B+T+F	0.660	0.920	+0.006	+0.013	0.599	0.826	+0.011	+0.018

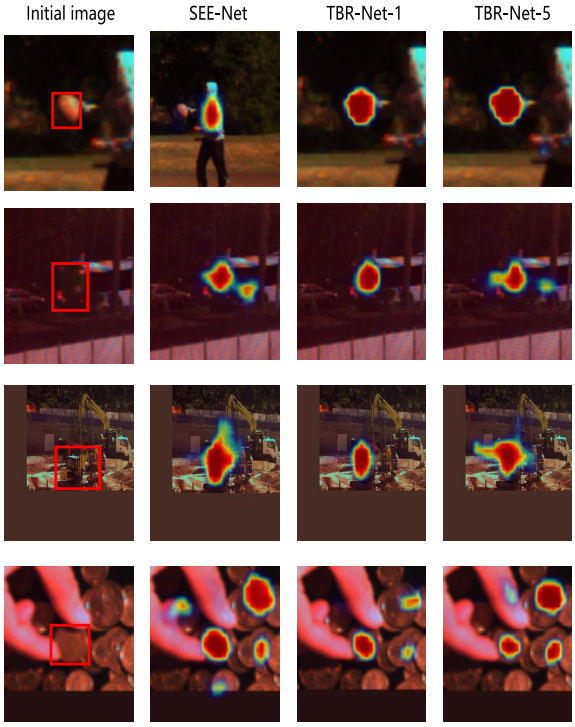

Fig. 9. Visualization of attention maps of baseline model and TBR-Net. The red color represents the high response, while the blue color represents the low response.

paper and *card* scenes, the RGB tracker struggles to handle the confusion between the object and the background, while the HS tracker can precisely locate the positions of the objects. Furthermore, TBR-Net offers better performance compared with the HS trackers, for example, frame 207 in the *paper* scene and frame 230 in the *forest2* scene. The visualization of tracking results confirms the advantages of our proposed methods.

b) Heatmap visualization: To enhance the credibility, we adopt the Grad-CAM [58] to visualize the attention map of the baseline model and TBR-Net. The results are shown in Fig. 9. First, the comparative experiment on the false-color image with the highest weight generated by the baseline model and TBR-Net is conducted. As shown in the second and third column of Fig. 9, TBR-Net has a better response to the object, proving that TBR-Net can capture richer semantic information related to the object. In addition, a comparative experiment on false-color images generated by TBR-Net with the highest and lowest weights was conducted. As shown in the third and fourth column of Fig. 9, it can be seen that the false-color image with the highest weight generated by the TBR method can produce a higher response to the object, proving that the weights generated by our TBR method can accurately reflect the importance of false color images to tracking performance.

C. Ablation Experiments

In this section, several ablation experiments on the HOTC dataset are conducted to show the effectiveness of our methods.

1) Evaluation of Each Component: The TBR-Net consists of three components: the Siamese backbone network, the proposed TBR, and FRM. We choose the SEE-Net [19] as our baseline (denoted as "B"), which adopts the SiamBAN [36] as the Siamese backbone, incorporating an SE module to regroup bands. The TBR-Net replaces the SE module with TBR, which is denoted as "T." Then, the FRM module (denoted as "F") is added to the network to form the completed model.

The quantitative comparison results are listed in Table VI. The baseline and the proposed methods regroup the HSI into multiple three-channel false-color images. We can observe that with the knowledge of RGB-domain Siamese backbone network, TBR improves the AUC score and DP_20 by 0.001 and 0.011, respectively, compared with the baseline model. It demonstrates the advantages of the global spectral context modeling of TBR, and the Transformer-based approach is better at handling the band regrouping problem than the SSC-based approaches. In addition, we further conducted ablation experiments on a broader dataset, HOTC-2023. We can see that the TBR-Net improves the AUC score and DP_20 by 0.011 and 0.018, respectively, which also proves the effectiveness of the proposed methods. The attribute-based comparison experiments are performed, and the results of precision plots are listed in Table VII. We can observe that in the IV, LR, and OCC scenarios, the performance of our method is higher than the baseline method. Besides, although the baseline model works better in some scenarios, TBR-Net outperforms the baseline model according to the average performance of the 11 attributes.

2) Evaluation of Different Embedding Strategy: To evaluate the effect of the embedding strategy in FRM, several experiments are conducted to explore better settings. Specifically, the patch level with different pooling sizes and the pixel level with different linear layers are used. As shown in Table VIII, the pixel-level embedding strategy with different linear layers performs better. This is because the linear transformation can convey information about the object across T & S features [35].

3) Evaluation of Different Band Regrouping Methods: To show the advantages of the proposed regrouping method TBR, the TBR is compared with other state-of-the-art band regroup

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TABLE VII

ABLATION EXPERIMENTAL RESULTS (DP_20) OF EACH COMPONENT. WE USE THE RED AND BLUE FONTS TO HIGHLIGHT THE TOP TWO VALUES

Method	BC	DEF	FM	IV	IPR	LR	MB	OCC	OPR	OV	SV	Average
B	0.943	0.928	0.867	0.873	0.963	0.857	0.855	0.846	0.961	0.860	0.914	0.897
B+T	0.930	0.922	0.949	0.879	0.953	0.944	0.982	0.873	0.947	0.860	0.911	0.922
B+T+F	0.934	0.920	0.946	0.885	0.952	0.944	0.983	0.876	0.948	0.864	0.912	0.924

TABLE VIII

ABLATION EXPERIMENTAL RESULTS (AUC) OF DIFFERENT EMBEDDING STRATEGIES. Δ INDICATES THE AMOUNT OF VARIATION

Methods	AUC	DP_20	Δ(AUC)	Δ(DP_20)
3×3 pool	0.641	0.903	-	-
7×7 pool	0.650	0.916	+0.009	+0.013
linear embedding	0.660	0.920	+0.019	+0.017

TABLE IX

ABLATION EXPERIMENTAL RESULTS OF BAND REGROUPING STRATEGY. Δ INDICATES THE AMOUNT OF VARIATION

Methods	AUC	DP_20	Δ(AUC)	Δ(DP_20)
BA	0.649	0.903	-	-
SE	0.655	0.907	+0.006	+0.004
TBR	0.654	0.918	+0.005	+0.015

TABLE X

ABLATION EXPERIMENTAL RESULTS OF DIFFERENT TRANSFORMER SETTINGS

Settings		Performacne			
Depth	Head	AUC	DP_20	Δ(AUC)	Δ(DP_20)
1	1	0.660	0.920	-	-
1	4	0.646	0.911	-0.014	-0.009
1	8	0.638	0.904	-0.022	-0.016
4	1	0.629	0.887	-0.031	-0.033
4	4	0.655	0.914	-0.005	-0.006
4	8	0.644	0.909	-0.016	-0.011
8	1	0.639	0.899	-0.021	-0.021
8	4	0.646	0.911	-0.014	-0.009
8	8	0.647	0.913	-0.013	-0.007

methods, including the SE module [16] and the band attention-aware (BA) network [19]. The three-channel false-color images are generated using the above three band regrouping methods, respectively, and we select SiamBAN [36] as the baseline model for feature extraction. All the parameters are kept consistent during the experiments. The evaluated results are listed in Table IX. As for the AUC score, the SE module and the TBR outperform the BA network. We can observe that using the TBR provides higher DP_20 than the other two comparative methods. The above experimental results further show the effectiveness of the proposed methods.

4) Evaluation of Different Transformer Settings: The depth and the number of heads are related to the amount of calculations in multihead attention. To explore the parameter setting of TBR, three depth configurations and the number of heads are used. The evaluated results are listed in Table X. The TBR with 1 head and 1 depth yields the better performance. Because the Transformer-based module is adopted as an "attention mask," and the Transformer encoder can capture the latent context information, the multihead attention calculation may lead to information confusion without the Transformer decoder.

V. CONCLUSION

In this article, we propose a TBR method and feature refinement network for object tracking in HSVs. The former captures the spectral context information and regroups the bands into several three-channel images. In this way, the visual confusion from BC can be resolved, and the prior knowledge from the existing RGB-domain trackers can be used to tackle the HS training data scarcity. The latter deals with the similarity modeling between pixels of T & S frames concerning the ROI focus property. Consequently, the correlation layer concentrates on the similar regions between T & S frames, further enhancing the tracking results. Experimental results compared with state-of-the-art trackers on the HOTC benchmark demonstrate the superiority of the proposed methods.

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